

SLIP – 18

Section I: Data Structures [15 Marks]

Q1) Sort a random array of n integers (accept the value of n from user) in ascending order by using bubble sort algorithm.

Answer:

```
#include <stdio.h> #include
<stdlib.h>

int main()
{
    int a[100], n, i, j, temp;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
    }
    return 0;

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp
                a[j] = a[j
+ 1];
                a[j + 1] =
temp;
            }
        }
    }
}
```

```
    printf("\nSorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database(Primary keys are underlined)

Student (rno,name,city)

Teacher(tno,tname,phone_no,salary)

The relationship is as follows:

Student-Teacher: Many-to-Many with subject as a descriptive attribute.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Student(
rno INT PRIMARY KEY,
name VARCHAR(50),      city
VARCHAR(50)
);

CREATE TABLE Teacher(
tno INT PRIMARY KEY,
tname VARCHAR(50),
phone_no VARCHAR(15),
salary NUMERIC(10,2)
);

CREATE TABLE Student_Teacher(      rno
INT REFERENCES Student(rno),      tno
INT REFERENCES Teacher(tno),
subject VARCHAR(50),
PRIMARY KEY(rno, tno, subject)
);

INSERT INTO Student VALUES
```

```
(1, 'Amit', 'Ahilyanagar'),  
(2, 'Neha', 'Sangamner'), (3, 'Raj', 'Pune');
```

```
INSERT INTO Teacher VALUES  
(1, 'Mr. Patil', '9822000001', 45000),  
(2, 'Prof. Patil', '9822000002', 55000),  
(3, 'Dr. Mulay', '9834000003', 65000),  
(4, 'Dr.Wani', '9834000004', 35000);
```

```
INSERT INTO Student_Teacher VALUES  
(1, 1, 'DBMS'),  
(2, 2, 'IKS'),  
(3, 3, 'Electronics'),  
(1, 4, 'IKS'),  
(2, 1, 'C');
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. List the students from “Sangamner” city.

Answer:

```
SELECT * FROM Student  
WHERE city = 'Sangamner';
```

ii. Find the maximum salary of teachers.

Answer:

```
SELECT * FROM Teacher  
ORDER BY tname;
```

iii. Count the number of students who have taken subject “IKS”.

Answer:

```
SELECT COUNT(DISTINCT rno)  
FROM Student_Teacher  
WHERE subject = 'IKS';
```

Q 2(C)

C. Write a query to list the names of all students who are taught by teacher “Dr.Wani” [4 marks]

Answer:

```
SELECT DISTINCT S.name  
FROM Student S  
JOIN Student_Teacher ST ON S.rno = ST.rno  
JOIN Teacher T ON T.tno = ST.tno  
WHERE T.tname = 'Dr.Wani';
```

Q 3) Viva [5 marks]